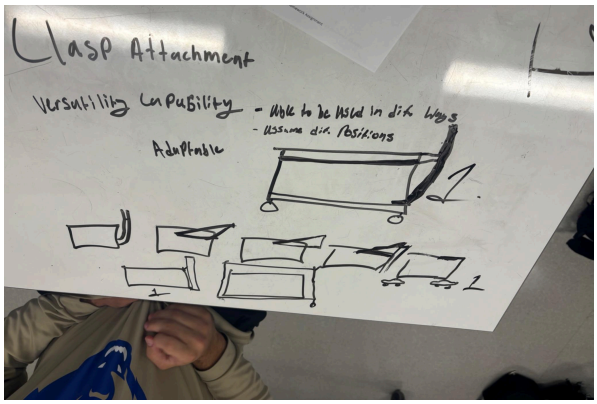
Element D

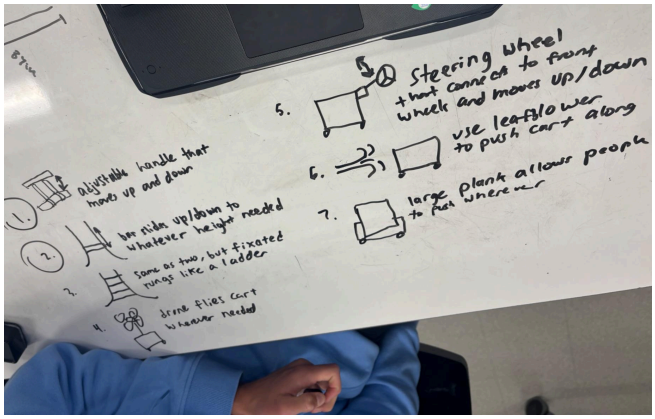
BRAINSTORMING

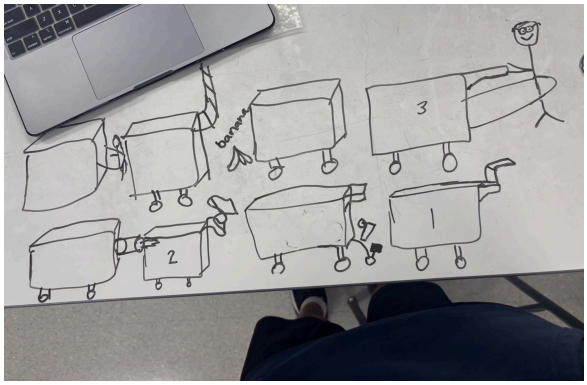
Picture 1: Nic ideas

Picture 2: Bennett ideas

Picture 3: Connor ideas







First, our team brainstormed multiple ideas separately. After we had each come up with about eight or so ideas, we collaborated as a team to choose three good options. We then created a design matrix where we compared six designs against criteria and constraints that were weighted by importance. We determined our criteria when we had a

conversation with our stakeholder, Mr. Kiser. He explained some qualities that he wanted in a good project, including adjustability, ergonomically fit, removability, and more as shown below. We took the highest scoring number and tried to make a prototype using that option as a basis.

CRITERIA WITH WEIGHT

Adjustable	0.33						
Ergonomically Fit	0.1						
Long-lasting	0.07						
Easily removable	0.3						
Functionality	0.14						
Budget	0.06						
sum = 1	1						
ALL TEAM MEMBERS RATED OUR SIX DESIGNS							
vote (e.g. use -1, 0, 1 scale)		**Scale is (-4,4)**					
Connor Harris	Weight as decimal	BPT	Towers	Triangular Bracing	Dolly	Rocket Launcher	Steering Wheel
Adjustable	0.33	4	0	-2	-3	-4	3
Ergonomically Fit	0.1	3	2	2	4	-2	3
Long-lasting	0.07	3	3	3	3	-3	2
Easily removable	0.3	2	3	-2	-4	4	4
Functionality	0.14	4	4	2	3	3	3
Budget	0.06	4	2	4	3	-4	2
	total score	3.23	1.99	-0.33	-0.98	-0.35	3.17
Nic Blucher	Weight as decimal	BPT	Towers	Triangular Bracing	Dolly	Rocket Launcher	Steering Wheel

Adjustable	0.33	4	0	-2	-4	-4	-3
Ergonomically Fit	0.1	1	3	2	2	-4	4
Long-lasting	0.07	-3	3	4	2	2	1
Easily removable	0.3	2	3	-4	-1	-3	-3
Functionality	0.14	2	3	2	3	-2	2
Budget	0.06	1	-1	-3	2	-4	-3
	total score	2.15	1.77	-1.28	-0.74	-3	-1.32
Bennett Lohmeyer	Weight as decimal	BPT	Towers	Triangular Bracing	Dolly	Rocket Launcher	Steering Wheel
Adjustable	0.33	4	-1	-1	-3	-1	2
Ergonomically Fit	0.1	2	3	1	2	-2	4
Long-lasting	0.07	2	3	3	4	0	-1
Easily removable	0.3	3	4	-1	0	-4	-2
Functionality	0.14	4	1	1	2	0	3
Budget	0.06	3	3	-1	1	-4	-3
	total score	3.3	1.7	-0.24	-0.17	-1.97	0.63
FINAL SCORES							
		BPT	Towers	Triangular Bracing	Dolly	Rocket Launcher	Steering Wheel
	total team scores	8.68	5.46	-1.85	-1.89	-5.32	2.48

FINAL DESIGN SELECTION →

This photo depicts a design using our highest-rated prototype idea that allows the handle to be increased in height. This was the first thing that we created and we will eventually add on to this to create our final prototype.

We also created our second-highest score in the design matrix just so we would have two options to choose from. We let Mr. Kiser (our Stakeholder) test both and give us feedback on how each of them feels. We had more problems with the design below because the wood near the bottom that we used as a clip onto the cart broke off too easily (it broke two times in the span of two or three weeks). In the end, we decided that our first of these two options would be the best one to continue making a final prototype on.